

Retail Sales Analytics using PySpark

Section 1: Project Overview

This project analyzes the Superstore Retail Dataset using PySpark and Spark SQL to evaluate sales performance, profitability, customer behavior, product performance, and regional trends. The objective is to provide actionable business insights that help management improve revenue, profitability, inventory planning, and customer retention strategies. The analysis includes data validation, KPI calculations, regional analysis, category analysis, customer analysis, product analysis, trend analysis, and business recommendations.

Section 2 – Data Validation

1. Schema Validation

The dataset contains 21 columns including order details, customer information, regional information, product information, sales metrics, discounts, and profit fields. Initially, all columns were loaded as string data types from the CSV file and required conversion for analytical processing.

2. Null Value Analysis

A column-wise null value check was performed across all fields.

Findings

- No significant null values were identified in the critical business columns such as:
 - Order ID
 - Order Date
 - Customer ID
 - Customer Name
 - Region
 - Category
 - Sales
 - Profit

The dataset was considered sufficiently complete for business analysis.

3. Duplicate Record Analysis

A duplicate record validation was performed using:

```
df_store.count() - df_store.dropDuplicates().count()
```

Findings

- No duplicate records were detected in the dataset.
- Each row represents a unique order-product transaction.

4. Data Type Validation

The dataset was initially imported with most columns as string types. Required conversions were performed:

Column	Original Type	Converted Type
Order Date	String	Date
Ship Date	String	Date
Sales	String	Double
Profit	String	Double
Discount	String	Double
Quantity	String	Integer

This ensured accurate aggregations and calculations during analysis.

5. Date Format Validation

During processing, inconsistent date representations were observed:

Examples visible in the dataset:

11/8/2016

10/11/2015

6/9/2014

and later converted into standardized Spark Date format:

2016-03-20

2015-09-18

2017-11-18

This conversion enabled monthly trend analysis and time-series reporting.

6. Numeric Format Validation

An invalid cast error occurred during SQL analysis when numeric fields were treated as strings:

CAST_INVALID_INPUT

This indicated that Sales, Profit, Discount, and Quantity fields required explicit type

conversion before aggregation and reporting. After conversion, calculations executed successfully.

7. Data Quality Conclusion

The dataset was found to be suitable for business analysis after:

- Converting date columns into proper date format.
- Converting numeric fields into appropriate numeric data types.
- Validating for null values.
- Checking duplicate records.
- Resolving data type inconsistencies causing casting errors.

The cleaned dataset was then used for KPI, regional, customer, product, and trend analysis.

Section 3: KPI Results

KPI	Description
Total Sales	Overall revenue generated by the business
Total Profit	Overall profit earned
Total Orders	Number of orders placed
Total Customers	Number of unique customers
Average Order Value	Average sales generated per order
Average Profit Per Order	Average profit earned per order

Key Observations

- The business generated strong overall sales revenue.
- Profitability is lower than revenue growth due to discount-heavy products.
- Customer base is diversified across Consumer, Corporate, and Home Office segments.
- A small percentage of customers contribute a significant portion of sales.

Section 4: Analysis Results

A. Regional Analysis

Best Performing Region

- West Region generated the highest sales and profit.

Worst Performing Region

- South Region generated comparatively lower sales and profit.

Findings

- Regional performance varies significantly.
- West and East regions contribute the majority of revenue.
- Some regions generate strong sales but lower profit margins.

B. Category Analysis

Most Profitable Category

- Technology

Least Profitable Category

- Furniture

Findings

- Technology provides the highest profit margins.
- Office Supplies contributes stable revenue.
- Furniture generates high revenue but lower profit due to discounting and product costs.

C. Customer Analysis

Top Customers

- A small number of customers contribute a significant share of total revenue.

Customer Segments

- Consumer segment represents the largest customer group.
- Corporate customers contribute substantial revenue.
- Home Office segment contributes the least.

VIP Customers

- Top 10 customers based on total sales were identified as VIP customers.

D. Product Analysis

Top Performing Products

- Several products consistently generate high sales and profit.

Loss-Making Products

- Certain products generate negative profit despite strong sales.

Findings

- High-selling products are not always highly profitable.

- Some products require pricing review due to continuous losses.

E. Trend Analysis

Sales Trend

- Sales show seasonal fluctuations throughout the year.

Profit Trend

- Profit growth does not always match sales growth.

Best Sales Month

- Peak sales occur during specific seasonal periods.

Worst Sales Month

- Off-season periods show reduced sales and profitability.

Section 5: Business Insights

1. West Region is the highest revenue-generating region and contributes significantly to overall business performance.
2. South Region is the lowest-performing region and requires additional growth initiatives.
3. Technology is the most profitable category and delivers the highest return on investment.
4. Furniture generates strong revenue but lower profit margins compared to Technology.
5. Office Supplies provides stable and consistent revenue across all customer segments.
6. A small group of VIP customers contributes a large percentage of total sales revenue.
7. Certain products generate high sales but negative profit, indicating ineffective pricing strategies.
8. Top-selling products contribute a significant portion of total revenue and should remain inventory priorities.
9. Monthly sales patterns reveal seasonal demand fluctuations.
10. Higher sales do not always translate into higher profits because of discount-related margin reductions.
11. Consumer and Corporate segments contribute the majority of total sales.
12. Profitability varies significantly across categories, regions, and products, highlighting opportunities for optimization.

Section 6: Recommendations

1. Increase marketing investment in high-performing regions such as the West to maximize revenue growth.
2. Review discount strategies for loss-making products, especially within the Furniture category.
3. Promote high-profit Technology products through targeted marketing campaigns.
4. Implement customer loyalty programs focused on VIP customers to improve retention.
5. Reduce inventory levels for consistently loss-making products and prioritize profitable alternatives.
6. Increase inventory availability for top-selling products to avoid stock shortages.
7. Develop region-specific marketing strategies to improve performance in underperforming regions.
8. Monitor profit trends alongside sales trends to ensure sustainable business growth.
9. Use customer segmentation to create personalized promotions for Consumer and Corporate customers.
10. Conduct regular product profitability reviews to identify products requiring price adjustments or discontinuation.

Conclusion

The analysis reveals that revenue growth is primarily driven by the West region, Technology category, and a small group of high-value customers. While overall sales performance is strong, profitability can be improved by optimizing discount strategies, focusing on high-margin products, strengthening customer retention programs, and addressing loss-making products. These actions will help the company achieve sustainable growth and improve overall business performance.